

BLAKE RAIN

I am a UK-based software engineer with 20 years of experience across FinTech, PropTech, film post-production, and industrial automation. I build across the stack: systems at the boundary of software and hardware (JIT compilers, packet processors, custom NIC drivers) and cloud-native distributed systems on AWS, from serverless orchestration to AI/ML integration. I usually work in Rust, C, or Haskell. Experienced in regulated environments including FCA and PCI-DSS compliance.

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SKILLS

LANGUAGES	Rust C++ C Haskell Python JavaScript TypeScript Go Bash SQL
SYSTEMS	Linux (embedded, drivers) macOS BSD
NETWORKING	Ethernet TCP/IP Packet Processing DPI SIP/RTP
COMPLIANCE	GDRP PCI-DSS FCA
DOMAINS	JIT/Compiler Garbage Collection Phone Systems (VoIP) Financial Modelling Embedded (ARM, FPGA) Graphics Film Post-Production
CLOUD	AWS (CF, Lambda, API Gateway, Bedrock)

EXPERIENCE

Cignpost Group ▶ Software Architect

Mar 2024 – Jul 2026

Financial Services / Health Insurance · Peper Harow, Godalming, UK · Fully remote

I was brought in to Cignpost as a Software Architect. I built a compliance operations platform for call transcription with AI analysis, flexible assessment workflows, task management, and staff training. I built an OAuth 2.0 SSO service for the health policy platform, mobile app, and internal services. I performed data analytics and mining on legacy portfolios to accelerate B2B and B2C growth.

Software Architect. I built the compliance operations platform, an OAuth 2.0 authorisation service, and performed data analytics and mining to support the health diagnostics product.

- Built a call compliance platform integrating Amazon Transcribe and Bedrock to generate transcripts with AI-flagged key moments, with an interactive player syncing audio, transcript, and annotations for FCA-mandated reviews. Built using Rust, HTMX, and Web Components.
- Designed a flexible assessments engine with configurable sections and multi-type question fields, enabling the compliance team to evolve their audit criteria without code changes.
- Developed a compliance task system (issue-tracker model) that unified findings from call assessments and email triage into actionable items with custom dispatch workflows.
- Created a training and examination portal: configurable exams with scoring, question-level feedback, and completion tracking across the workforce.
- Built an OAuth 2.0 authorisation service (RFC 6749, 7636, 7662, PASETO) for SSO across the health policy platform and mobile app with TOTP support and Lua policies. Included a management API and an HTMX admin portal.
- Performed data analytics and mining on legacy insurance portfolios to accelerate B2B and B2C growth for the health diagnostics product.

Rust PostgreSQL LLMs HTMX AWS (CF, S3, Lambda, Transcribe, Bedrock) OAuth 2.0
TOTP PASETO TTS

Neo Technologies Limited ▶ Technical Director

Jan 2019 – Jan 2024

Telecoms · London, UK · Hybrid (one day per week in office, full-time during audits)

Eclipse was a PCI DSS descoping device: a layer-2 transparent appliance that sat inline in a contact centre's network, passively inspected SIP traffic, and redacted DTMF tones and audio from media streams during payment capture. Because the device was not directly addressable and became the entire cardholder data environment, it removed almost all PCI DSS scope from both the customer and from Neo itself. The product later pivoted into a general-purpose SDN prototype.

Technical Director and lead engineer. I designed and built the device firmware, the cloud orchestration infrastructure, and the agent-facing control panel. I owned the technical aspects of the PCI DSS Level 1 Service Provider compliance programme

(architecture, technical controls, and audit engineering) working alongside the MD on operational, risk, and continuity requirements.

- Designed a layer-2 transparent appliance that acted as the entire cardholder data environment, removing PCI DSS scope from contact centre infrastructure.
- Built a JIT-compiled packet processor (libjit, x86-64 and ARM) that dynamically rewrote packet filtering rules per SIP session, enabling real-time DTMF redaction and audio silence injection on multi-Gbps traffic.
- Wrote a custom user-space driver for Intel Gigabit Ethernet controllers (i350 family) with a packet buffer garbage collector shared with Rust via DMA-aware memory layout.
- Designed a WebAssembly gateway interface: payment modules were signed, sandboxed Wasm binaries that could build HTTP requests but had no direct network access.
- Built AWS Lambda-based cloud orchestration that brokered capture sessions between devices and agent panels, handled administration, and maintained full audit trails.
- Developed an agent control panel using Rust/Wasm (Yew + Tauri) that ran as a standalone desktop app, a web app, or an embedded iframe in existing CRMs.
- Achieved and maintained PCI DSS Level 1 Service Provider certification with zero findings across multiple annual audits, and transitioned from 3.2.1 to 4.0.
- Deployed Eclipse in a production data centre handling 8-10 TB of daily traffic across multiple large-scale contact centres.
- Pivoted the core packet-processing engine into an SDN prototype, with signed Wasm modules granted direct control over the JIT packet processor.
- Solo-engineered the product post-COVID after the engineering team was made redundant during lockdown.

Rust C++ JIT x86-64/ARM AWS (CF, Lambda, API Gateway, Compliance) WebAssembly
HTMX PostgreSQL PCI DSS

Inchora Limited ► Head of Technology/Research

Sep 2014 – Jan 2019

PropTech / Financial Services · Farnborough, UK · Remote (with regular visits to Farnborough, London, and Grantham)

The Inchora Group combined businesses across the UK lettings industry and regulated financial services (mortgages, life insurance). I led technology across two major systems: Inchora Home (tenant utility notifications for ~1,700 letting agent offices) and an internal contact centre CRM (~100 agents) that sold insurance, media, and energy products. The platform managed ~1.4m tenants at exit.

Head of Technology. I architected and built the core systems: the CRM, dialler, campaign engine, agent desktop, and tenant notification platform. I grew the engineering team to five members. I also worked with the CTO to manage PCI DSS compliance across the group.

- Designed a campaign graph engine that compiled management-authored visual graphs into optimized SQL, powering automated outreach (SMS/email), dial queue insertion, and event-driven triggers.

- Built a dynamic dial-queue system where lead priority re-ordered in real time based on configurable properties (such as move-in proximity, sales history, overdue status), with a dispatcher balancing fairness against throughput and intelligent transfer routing.
- Developed an Electron-based agent desktop with TAPI phone integration, handling multi-category sales journeys.
- Architected a scripting layer (JavaScript) for customer APIs and event processing, enabling multiple third-party integrations without deployment cycles.
- Maintained PCI DSS compliance across the group.

Haskell

PostgreSQL

JavaScript/TypeScript

React

Electron

AWS

AmberVisual Limited ► Software Engineer

2005 – 2008

Film post-production · Huddersfield, UK · Hybrid (on-site early, remote later)

Halide (named for silver halide film chemistry) was a film-look tool for digital footage post-production. It let editors apply a configurable stack of effects such as colour mapping and convolution filters. Effects were authored as visual graphs by our in-house specialist. The product shipped as an Adobe AfterEffects plugin and a headless render-farm pipeline, and was demoed at NAB Las Vegas.

Sole engineer on Halide. I designed and built the runtime effects engine with its JIT compiler, the deinterlacer, and the in-house node editor used to author the effects.

- Built a domain-specific JIT compiler that transformed effect graphs into optimized machine code, enabling real-time HD playback on consumer hardware (Athlon64, SSE2/3).
- Wrote a custom instruction selector with a CPU pipeline model that maximized instruction tiling for specific processor architectures.
- Developed a deinterlacer using temporal/spatial hysteresis motion masks and edge-vector reconstruction, matching the market-leading competitor in quality.
- Built a GTK-based node editor that let non-programmers visually design effect pipelines, add parameterization, and export as deployable bundles.
- Shipped Halide as an interactive AfterEffects plugin and a headless render-farm version.

C++

GTK

SIMD (SSE2/SSE3)

EARLIER CONTRACTING

Industrial Automation — Obsidian (Leeds startup)

2004

Worked across multiple engagements for a Leeds-based industrial automation startup. Contributed to flowchart-based tooling, a distributed consensus architecture for field units, a custom FPGA-driven graphics pipeline, and a Linux base image for a GSM-connected vending machine product.

- Built a flowchart-based editor and evaluation engine for ARM field units communicating over Ethernet, with a distributed consensus mechanism to resolve state across devices
- Designed an FPGA-driven polygon span renderer with edge sorting to eliminate overdraw, receiving draw commands from the CPU and maintaining framebuffers in hardware, reducing UI update time from seconds to sub-millisecond
- Developed a custom Linux distribution for a GSM-connected vending machine, including kernel drivers for the dispenser hardware"

C ARM FPGA / Verilog Linux (custom kernel)

PropYield — Commercial Property Calculator

2012 - 2014

Built a SaaS calculator for commercial property investment analysis. The calculator performed goal-seeking across multiple financial variables (yield, LTV, equity return, interest cover, etc) using Powell's dog leg method. Included saved calculations, sharing, and PDF export.

- Implemented Powell's dog leg root-finding algorithm to solve for arbitrary financial targets, such as adjusting annual rent to hit a desired net yield, or solve for LTV given a required equity return
- Wrote a fast QR factorisation in C to replace a slow Haskell implementation, after profiling showed the numerical core was the bottleneck
- Delivered as a SaaS product with account management, saved calculations, sharing, and PDF export

Haskell C JavaScript